

JEROME FRANCIS

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EDUCATION

Master of Science in Computer Science (Machine Learning), 2023 – 2025 | Montreal, Canada
Mila - Quebec AI Institute / Université de Montréal (3.8 / 4.3 GPA)

Bachelor of Technology (Honours) in Computer Science, 2016 – 2020 | Kollam, India
T.K.M. College of Engineering (9.3 / 10 GPA)

PROFESSIONAL EXPERIENCE

AI Research Engineer & Mentor (Remote), Algoverse AI Research 06/2025 – Present

- Led a 12+ member team building ML pipelines for **RAG bias mitigation, long-context LLM hallucination analysis**, improving SLM reasoning and **RL/GRPO post-training**. Defined experimental design, evaluation methodology, and analysis workflows to improve reproducibility and model benchmarking.
- Accelerated GPU workflows using **DeepSpeedZero, FSDP** frameworks, Mixed Precision Training and **CUDA** tensor optimization for distributed ML and **PEFT** (LoRa, Q-LoRa) training processes.
- Interpreting Latent CoT trajectories as Dynamical Systems - Accepted at **ICML 2026 FoGen Workshop** 

Machine Learning Engineer, Intern, Shearwater Aerospace 05/2024 – 01/2025 | Montreal, Canada

- Designed and implemented **scalable data ingestion** and processing pipelines for **700k** Leonardo flight logs using **GCP** and Google Earth Engine and **geospatial datasets** (ERA5, MODIS, ArcGIS) to create **reproducible ETL workflows** for training dataset at scale. Increased thermal prediction accuracy from **42% to 60%**.
- Built an end-to-end ML pipeline using **MLflow** with a **UNet segmentation** model for thermal localization, achieving a **0.66 Dice score** on validation and optimizing for deployment constraints.
- Collaborated in a **cross-functional** team of aeronautical and meteorological domain experts in autonomy software engineering and product team

Data Scientist, Tata Consultancy Services 2020 – 2023 | New Delhi, India

- Partnered with **risk management** teams to build and maintain **credit loss forecasting pipelines** using **XGBoost** for PD, LGD, EAD models, that improved reserve planning accuracy by **10%** and supported executive risk decisions.
- Modernized **18+ legacy** CECL/CCAR **pipelines** for **PNC Financial Services** into standardized, production-grade workflows, improving reliability, traceability, and execution performance
- Improved Non-Life insurance tariff pricing using GLMs, H-GLMs, and GAMs with enhanced credibility estimation, impacting **10+** products.
- Built distributed data pipelines (Apache Airflow + SQL + PySpark) to integrate data (GDPR) across **enterprise regulatory reporting** systems, reducing data preparation time by **33%**, delivering risk analysis and business intelligence for operational solutions with **7x TAT** at **Citibank NA**

Machine Learning Engineer, Intern, Ignitarium Technology Solutions 06/2019 – 07/2019 | Kochi, India

- Developed a real-time computer vision pipeline **UNet** segmentation pipeline using PyTorch for detecting faulty gaps (**anomaly detection**) on railway tracks, and curated and annotated **2,000** video frames across diverse terrains.
- Accelerated on-device inference by **5%** through **TensorRT optimization** and **ONNX model compression**, enabling deployment on the Rail Defect Detection Platform 

TECHNICAL PROJECTS

911Automate , RAG system for 911 EMS emergency protocols

Designed and built a production-ready **hybrid RAG** system combining dense retrieval (**Qdrant** vector DB) and **BM25**/sparse retrieval, with **cross-encoder reranking** and token-aware chunking. Implemented async FastAPI serving, CI-driven LLM-as-judge evaluation (**DeepEval**) to measure relevance/faithfulness in CI, with **prompt-engineering** and production-ready vector ingestion (OCR text, HIPPA) and pruning policies for latency/scale control. [Featured in *AI Tinkerers MTL*] 

User Notification Timing system for Recommendations 

Built a scalable **ETL** data and ML pipeline (**dbt** + **Airflow**) for **event-level modeling** and leakage-safe feature engineering. Trained a probabilistic engagement model with fatigue features and deployed online inference via model registry (MLflow), to improve user engagement. Included reliability and data quality checks (using **GreatExpectations, SHAP**, **LIME**).

QORTEX , 1st Place, Distinction QvStudio Prize at Mil'HaQ Fest 2025 by TD-Layer6 AI

Built a quantum-classical **price forecasting pipeline** using **Variational Quantum Autoencoder** & XGBoost/LightGBM in Qiskit and Pytorch to predict 14-day **swaption prices** for multiple contracts, achieving **0.96 R2** (validation).

PerplexMATH - Steering LLMs towards Math Reasoning , Poster accept at ICML 2025 NewinML workshop

Implemented a scalable evaluation benchmark and prompt-engineering suite on MATH-500 dataset using models like Deepseek, Llama3, Gemini, to measure LLM memorization, reasoning, and problem-solving, by using reference-based LLM as a judge evaluation, Analyzed compute vs. exploratory reasoning trade-offs.

Reinforcement Learning in Minecraft , 12th Position in NeurIPS MineRL Diamond challenge

Designed an agent in Minecraft to perform certain skills like chopping wood, crafting pickaxes, collect items like stones, food for survival using Policy Gradient technique (PPO) which got an average score of 19.0 per environment run (32 *environments*).

TECHNICAL SKILLS

Programming Languages: Python, SQL, NoSQL, HTML, CSS, Typescript, C++, R, LATEX, Linux

Frameworks: PyTorch, TensorFlow, Transformers, JAX, verl, FastAPI, Apache Spark, React | **Databases:** PostgreSQL, MySQL, MongoDB, Redis, BigQuery, Cassandra, Neo4j GraphDB

Tools and Platforms: HuggingFace, Docker, Kubernetes, GCP, Vertex AI, AWS (Sagemaker, Bedrock, Lambda), S3, MLflow, dbt, DVC, AirFlow, Github Actions, PySpark, Kafka, Flink, Prometheus/ Grafana, Snowflake, **Databricks**, Slurm, **CUDA** 

Artificial Intelligence: Machine Learning, Statistics, Deep Learning, Computer Vision, Reinforcement Learning , NLP, Time Series Forecasting, **Predictive Modelling**, Anomaly Detection, Inference Optimization, Model Quantization, Distributed ML Training, **A/B testing**

GenAI / LLM: Langchain, Langgraph, Langsmith, vLLM, deepeval, promptfoo, Google ADK, Qdrant, Autogen, MCP, CrewAI, n8n, Cursor, Claude